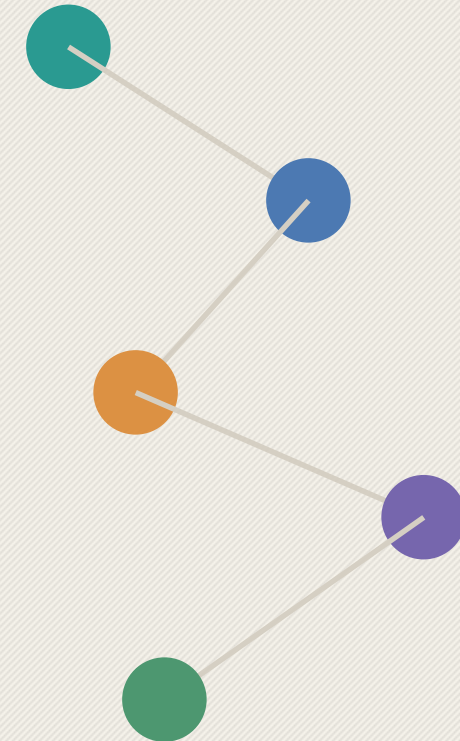


HackViz

Explainable, Transfer-Learning-Enabled Insider-Threat Detection

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Behaviour → Context → Evidence

The Problem

Why insider-threat detection needs behavioural context

From isolated events → behavioural patterns

- Static SIEM rules depend on predefined thresholds and patterns.
- Legitimate employees can naturally produce unusual-looking events.
- Sophisticated insider activity may not match a predefined rule.
- Raw logs are heterogeneous and lose meaning when treated as independent rows.
- Security teams need ranked, explainable alerts—not just an opaque risk score.

Goal: learn what “normal” looks like for each user over time.

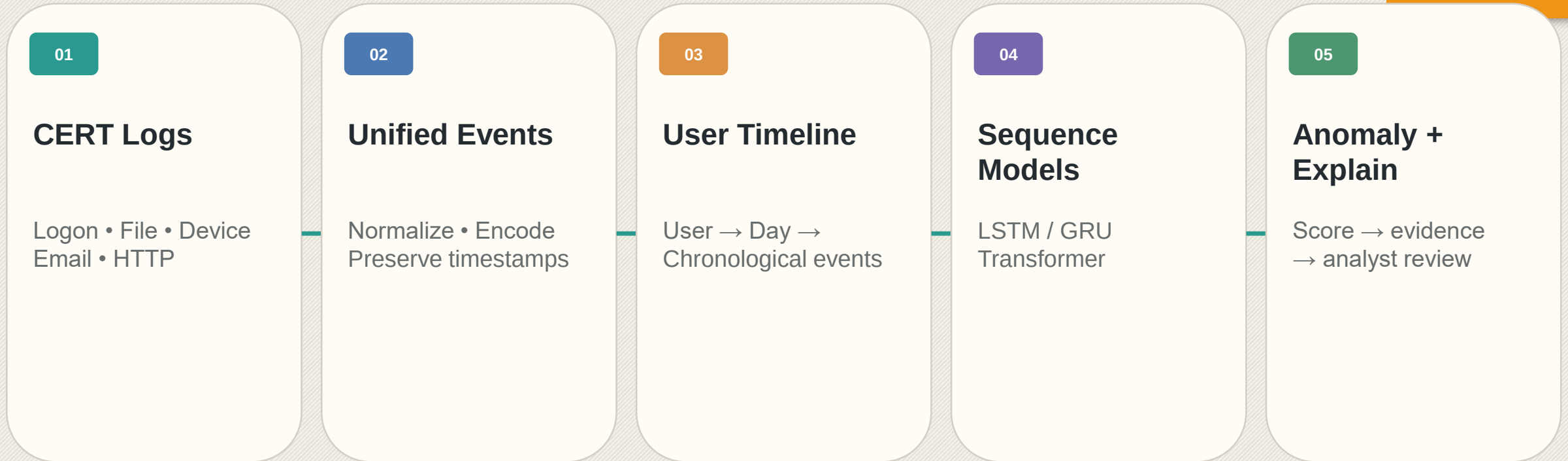
Typical challenge

08:55	●	Logon
09:12	●	File access
09:48	●	Email
11:06	●	HTTP
14:21	●	USB device

An event can be harmless alone, but suspicious in context.

HackViz at a Glance

One behavioural pipeline from raw activity to analyst-ready evidence



Design principle: the same preprocessing + temporal representation is shared across model variants for a fair comparison.

Heterogeneous Data Fusion

Turn five activity sources into one behavioural story

LOGON

Authentication activity

FILE

File operations

DEVICE

Device usage

EMAIL

Email activity

HTTP

Web activity

Unified behavioural timeline

USER U17

DAY 12



Sessionization + feature encoding → model-ready sequence

Labels stay separate and are used for evaluation, reducing leakage risk.

Multi-Architecture Comparison

Same input, same evaluation framework — different sequence modelling strategies

Baseline

LSTM / GRU

Learns local & short-range behavioural dependencies



Long-range

Transformer

Self-attention captures relationships across distant events



Combined

Hybrid LSTM + Attention

Order-sensitive recurrence + attention-based dependency modelling



The novelty is not “using LSTM” or “using Transformers” alone — it is the controlled empirical comparison on the same behavioural representation.

Transfer Learning for the Cold-Start Problem

Can prior behavioural knowledge reduce the history needed for a new environment?

CERT r4.2 → CERT r6.2

Generic model

**Local
adaptation**

Compare against training from scratch

Day 5

Day 10

Day 20

amount of user history available

Research question

How much behavioural history is enough to obtain useful detection after adaptation?

- Measure F1 and false-positive rate at each history level.
- Test transfer learning versus from-scratch training.
- Quantify whether cold-start effort can be reduced.

Analyst-Facing Explainability

Move from “anomaly score = high” to “these events made it suspicious”

Alert

User U17 • High anomaly

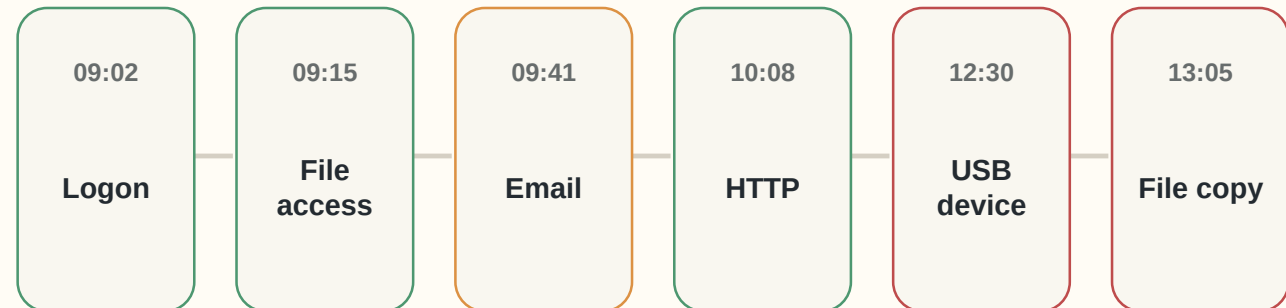
Anomaly score

0.92

Analyst evidence

- Unusual file activity
- Rare device usage
- Unexpected event sequence

Behavioural sequence — highlighted evidence



Attention information / highlighted events are presented as supporting evidence for human review — not as proof of malicious intent.

Current Status & Next Phase

Progress-report snapshot — what is completed, in progress, and planned



DONE

Problem + dataset selection

DONE

Requirements, use cases, DFDs, SRS, user stories

IN PROGRESS

CERT normalization + sessionization

NEXT

LSTM/GRU, Transformer + Hybrid models

NEXT

Transfer-learning Day 5 / 10 / 20 experiments

NEXT

Explainability + Streamlit analyst workflow

Next milestone → model implementation + controlled evaluation + dashboard integration